Shubham Gupta

Founding Engineer | Real-Time Voice AI | C++20 Systems | AI-Native Builder

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Summary

I build AI that talks back in real time. As founding engineer at Vaani AI Research, I built the C++20 engine behind its phone and web voice agents from scratch: sub-200 ms voice-to-voice, 21 AI vendors with live failover, built AI-native. Previously shipped connected-vehicle software for the New Renault Duster.

Experience

Vaani AI Research Bengaluru, India

Founding Engineer – Senior Software Engineer *May 2026 – Present*

- **Built the voice AI engine from zero:** a production C++20 orchestrator running live conversations over SIP telephony, WebRTC, and chat on LiveKit, streaming STT → turn detection → LLM → TTS in a sub-200 ms voice-to-voice budget.
- **Made live calls outage-proof:** router-based failover with PCM replay across 21 AI vendors (7 STT, 6 LLM, 8 TTS, incl. Deepgram, Sarvam, OpenAI, Gemini, Groq, self-hosted Gemma on vLLM, Cartesia, ElevenLabs), so no outage drops a call.
- **Made the agent feel human:** speculative LLM execution starts replying before the caller finishes, and generation-ID barge-in instantly kills stale tokens and audio the moment the caller interrupts.
- **Taught it when to speak:** multimodal end-of-turn detection fusing a text EOU classifier with the Smart-Turn audio model on ONNX Runtime / OpenVINO, plus VAD and neural noise suppression across 8/16/48 kHz audio.
- **Zero locks, zero allocations on the audio hot path:** lock-free ring buffers, strand-safe timers, and Boost.Asio, proven with RapidCheck property tests under ASan, UBSan, and TSan.
- **Built to extend, not edit:** ABI-stable stage contract, `dlopen` plugin host, and CI-enforced dependency lint; call recording, persona memory, and 12 session watchers shipped with zero core-engine changes.
- **15+ in-call tools via LLM function calling:** SIP warm/cold transfer, Cal.com booking, WhatsApp, RAG over pgvector, custom APIs, and voicemail detection.
- **Smarter conversations:** token-budgeted context with rolling summarization, persona memory from past calls and CRM, multilingual / Indic language switching, and guardrails.
- **Production-grade at scale:** actor-model agent server with Redis-backed concurrency and pooled connections on Kubernetes (HPA, zero-downtime deploys, 300 s graceful drain), with OpenTelemetry, Prometheus, and AWS / Azure secret managers.
- **AI-native engineering:** authored agent rules, `AGENTS.md`, Cursor skills (incl. a 4-phase architecture audit), and a 30K-node codebase knowledge graph that let coding agents ship production C++ safely.

Renault Group Bengaluru, India

Engineer *Jul 2025 – Apr 2026*

- **New Renault Duster:** built Android and embedded software for its connected-vehicle and in-vehicle apps (Kotlin, Java, Jetpack MVVM), owning two core components:
  - **DiagRef Handler:** vehicle diagnostics reference handling over microservices and REST APIs.
  - **Config Hub Bridge:** vehicle configuration sync between in-vehicle apps and backend services, with SQLite / Room persistence.
- **ADAS features:** developed driver-assistance features; prototyped Rust and OpenCV image processing for safety R&D.
- **Performance and team:** tuned for low latency with unit tests, code reviews, and CI/CD in Agile sprints; mentored new trainees.

Renault Group Bengaluru, India

Graduate Engineering Trainee *Jul 2024 – Jun 2025*

- Developed Android applications (Java, Kotlin, Android SDK) for connected-vehicle systems with SQLite and REST APIs.
- Improved reliability through debugging, unit testing, and bug fixes; worked with embedded Linux and automotive frameworks.

Technical Skills

Languages:	C++ (C++17/20), Kotlin, Java, Python, Rust, SQL, Bash
Systems:	Real-Time Audio, Multithreading, Lock-Free Data Structures, Boost.Asio, Actor Model, Plugin/ABI Design, CMake, vcpkg
Voice & AI:	LiveKit, WebRTC, SIP, STT, TTS, LLM Orchestration, Function Calling, RAG, pgvector, ONNX Runtime, OpenVINO, VAD, vLLM
Cloud & DevOps:	Docker, Kubernetes, AWS, Azure, Redis, PostgreSQL, GitLab CI/CD, OpenTelemetry, Prometheus
Mobile & Embedded:	Android SDK, Android Studio, Android Jetpack, MVVM, Room, SQLite, Gradle, Embedded Linux, OpenCV
Testing:	CTest, RapidCheck (Property-Based Testing), ASan, UBSan, TSan, clang-tidy, Unit Testing
AI Tooling:	Cursor, Claude, Agentic Coding, Prompt Engineering, Agent Rules & Skills, Codebase Knowledge Graphs
Domains:	Conversational Voice AI, Telephony, Connected Vehicles, ADAS, Vehicle Diagnostics, Agile / SDLC

Education

Chandigarh University Punjab, India

Bachelor of Engineering in Computer Science *2020 – 2024*